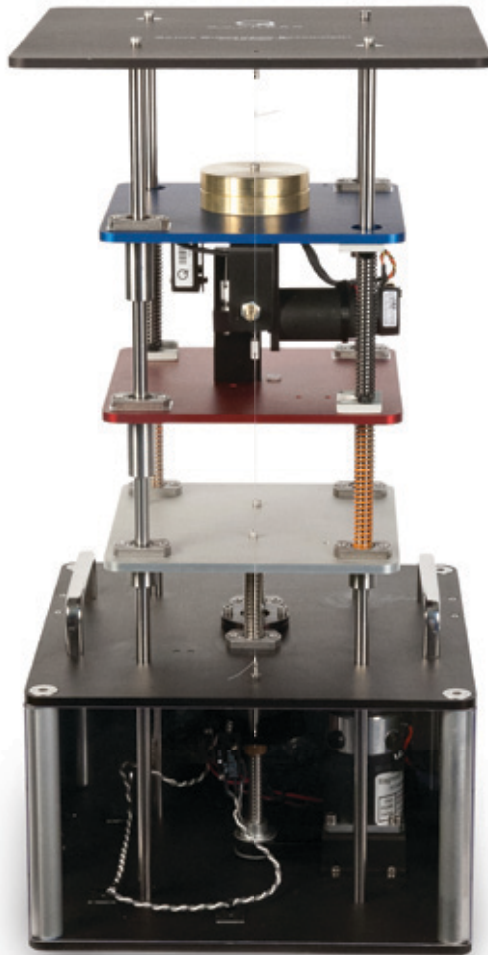




USER MANUAL

Active Suspension Experiment

Set Up and Configuration



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Quanser Inc.
119 Spy Court
Markham, Ontario
L3R 5H6
Canada
info@quanser.com
Phone: 1-905-940-3575
Fax: 1-905-940-3576

Printed in Markham, Ontario.

For more information on the solutions Quanser Inc. offers, please visit the web site at:
<http://www.quanser.com>

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1 PRESENTATION

The Active Suspension Experiment (ASE) comes complete with a laboratory Guide for undergraduate and graduate students and teaches a cutting-edge technology that has brought a new generation of vehicles to life. Conventionally, automotive suspension designs have been a compromise between the three paradoxical criterion of road handling, suspension travel and passenger comfort. Active suspension technology is used in the automotive industry to continuously control the vertical movement of the wheel using an actively controlled actuator placed on the suspension axis.

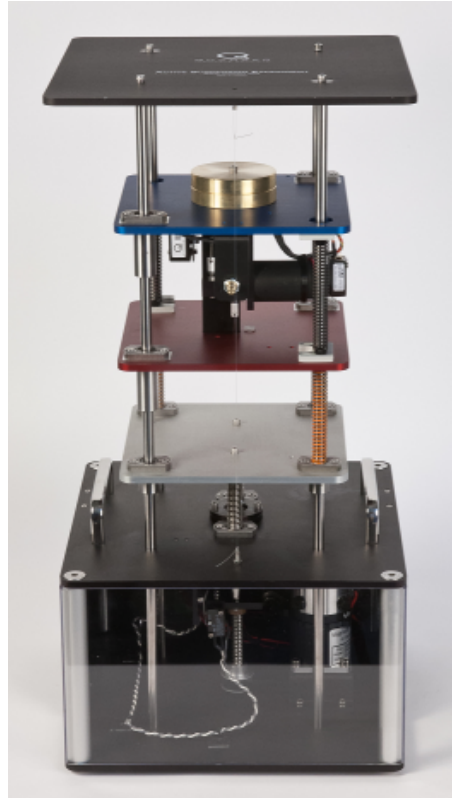


Figure 1.1: Quanser Active Suspension Experiment

In recent years the use of Active Suspension technology has allowed car manufacturers to achieve all three desired criterion independently. Similar technologies have also been used in train bogies to improve the curving behavior of the train and the decrease acceleration perceived by the passenger.

■ **Caution:** This equipment is designed to be used for educational and research purposes and is not intended for use by the general public. The user is responsible to ensure that the equipment will be used by technically qualified personnel only.

1.1 Active Suspension Experiment: General Description

The Active Suspension Experiment, ASE, is a bench-scale model to emulate a quarter-car model controlled by an Active Suspension mechanism. The plant consists of three floors/plates on top of each other. The top floor resembles the vehicle body and is suspended over the middle plate with two springs. A capstan drive high quality DC motor is also standing between the top and middle plates to emulate an active suspension mechanism. The top floor is instrumented with an accelerometer to measure the acceleration of the vehicle body relative to the plant ground. The middle plate is in contact with the bottom plate, i.e. the road, through a spring and constitutes the tire in the

quarter-car model. The bottom floor provides the road excitation in the system. It is connected to a fast response DC motor so that the designer can simulate different road profiles. When the motor turns, the torque created at the output shaft is translated, through the lead screw and gearing mechanism, to a linear force which results in the bottom plate's motion. The structure is made of steel and the three plates can smoothly slide along a stainless steel shaft using linear bearings. The motion of the two bottom plates is tracked directly by two high resolution optical encoders while a third encoder measures the motion of the top plate relative to the middle one. Such a scaled quarter-car structure has been designed to study critical aspects of Active Suspension control implementations.

1.2 Active Suspension Experiment: Control Challenge

The ASP consists of two masses, each supported by two springs as shown in the Figure 1. The upper mass represents the vehicle body supported above the suspension while the lower mass corresponds to one of the vehicles wheels. The entire system represents the classic quarter car model. An LQR controller can be used to optimize a variety of performance parameters in this fourth order quarter-car model. The performance criteria can be formulated into a mathematical model. This mathematical representation is then optimized while considering the control actuator limitations. The performance measures to be controlled by students include:

- Ride Comfort - is related to vehicle body motion sensed by the passengers. It is measurable through an accelerometer located on the sprung mass.
- Suspension Travel - refers to relative displacement between the vehicle body and the wheel and is constrained within an allowable range of motion. The relative displacement between the sprung mass and the unsprung mass represents Suspension Travel and is measured using a linear capstan mechanism.
- Road Handling - is associated with the contact forces between the road surface and the vehicle tires. The contact forces between the road and the tires depend on the tire deflection. In the Active Suspension structure the relative displacement between the unsprung mass and the road represents the tire deflection.

2 COMPONENTS

The Active Suspension Experiment components are identified in Section 2.1. Some of the those components are then described in Section 2.2.

2.1 ASE Component Nomenclature

The ASE components listed in Table 2.1 below are labeled in Figures 2.1, 2.2, 2.3.

ID	Component	ID	Component
1	Top Plate (Blue, Vehicle Body Mass)	2	Accelerometer Gain Potentiometer
3	Middle Plate (Red, Vehicle Tire Mass)	4	Suspension Encoder
5	Bottom Plate (White, Road)	6	Suspension Motor Capstan Cable
7	2 Adjustable Springs (Vehicle Suspension Springs)	8	Spring Holder Set Screw
9	2 Adjustable Springs (Vehicle Tire Springs)	10	Linear Bearing Blocks
11	Bottom Plate Encoder Connector	12	Top Plate Encoder Connector
13	Bottom Plate Counter Weight Springs	14	Payload Mass (Brass)
15	Active Suspension DC Motor	16	Bottom Plate Encoder
17	Bottom Plate Servo Motor	18	Lead Screw
19	Encoder Thread	20	Stainless Steel Shafts
21	Accelerometer	22	Accelerometer RCA Connector
23	Plant Top Cover	24	Limit Switch Safety Lights
25	Plant Handles	26	Bottom Plate Motor Connector
27	Limit Switch Push Key	28	Safety Rod
29	Movable Spring Holders	30	Safety Limit Switch
31	Suspension Motor Connector	32	Suspension Encoder Connector
33	Encoder Thread Anchor	34	Top Plate Encoder

Table 2.1: Active Suspension Experiment Components

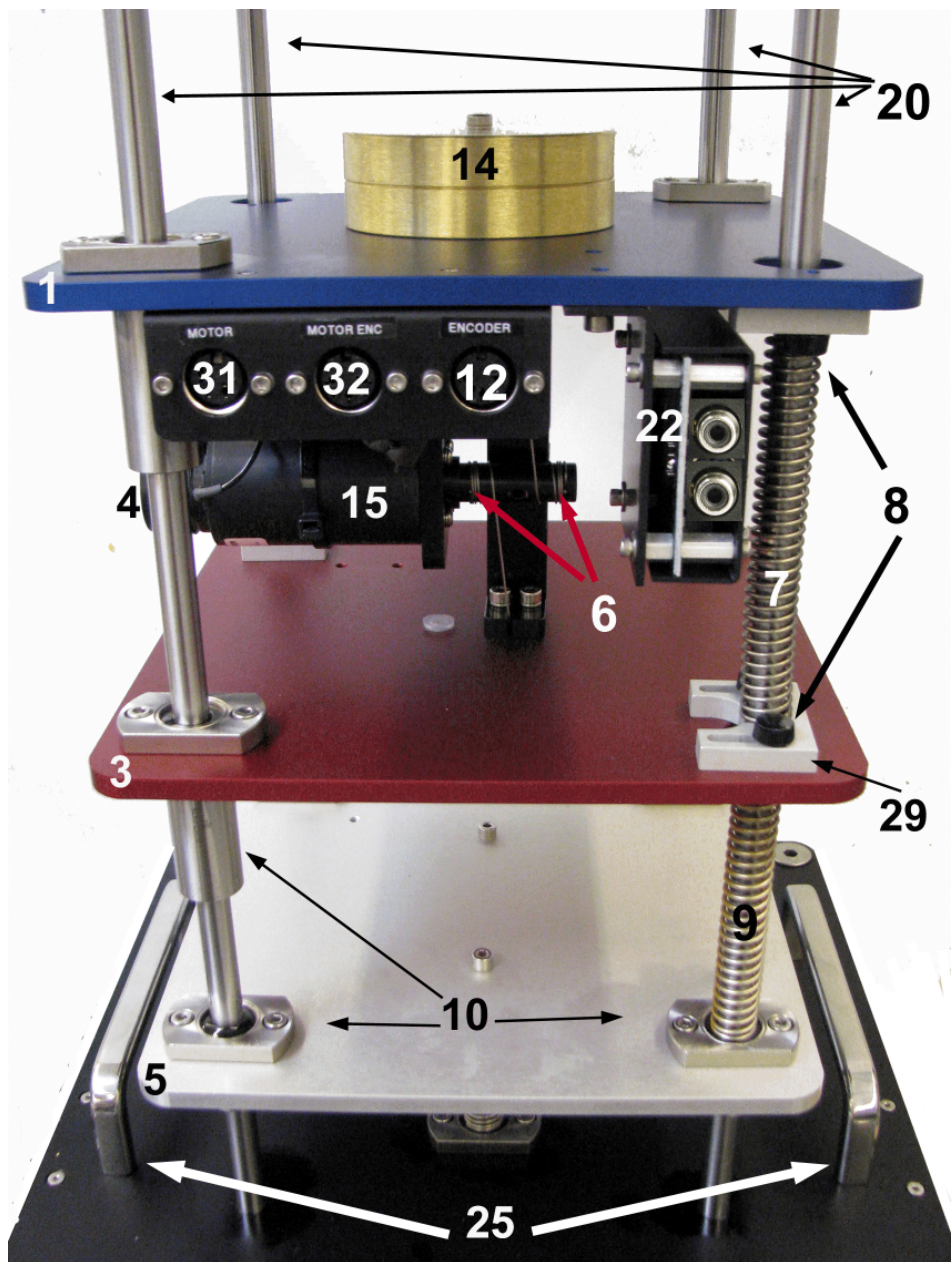
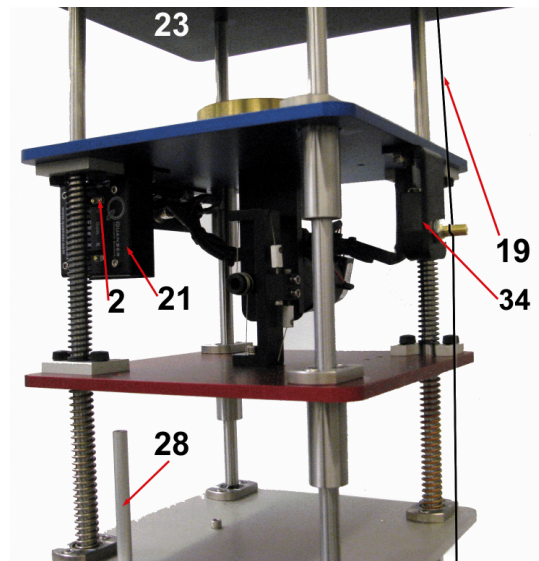


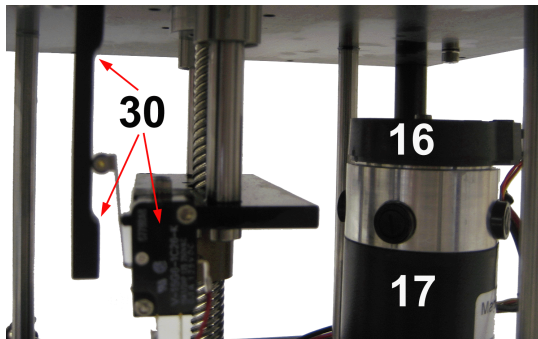
Figure 2.1: Active Suspension Experiment Top Front View



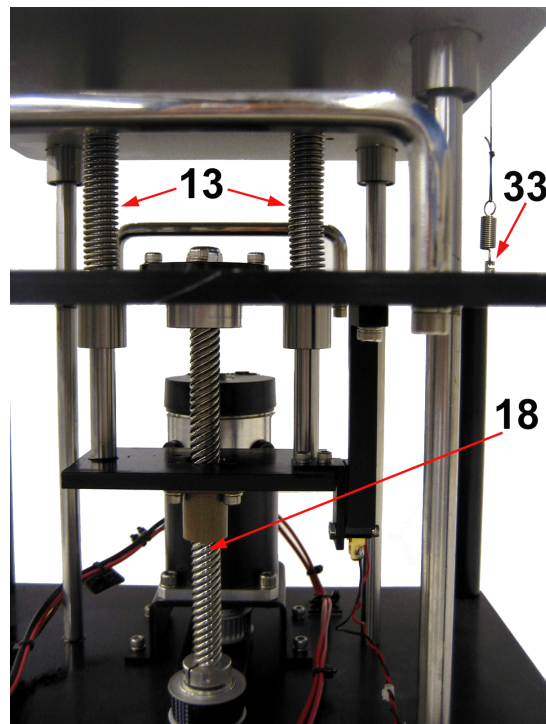
Figure 2.2: Active Suspension Experiment Bottom Front Panel



(a) Side View



(b) Base Inside View



(c) Base Inside View

Figure 2.3: Active Suspension Experiment Side View

2.2 Component Description

2.2.1 Active Suspension DC Motor (Component # 15)

The Active Suspension Experiment incorporates a custom **Faulhaber Brushless DC Motor** and is shown in Figure 2.1 with ID #15. This is a high efficiency, low inductance motor that can obtain a much faster response than a conventional dc motor.

■ **Caution:** *High-frequency signal applied to a motor will eventually damage the gearbox motor and the motor brushes.* The most likely source for high frequency noise is derivative feedback. If the derivative gain is set too high, a noisy voltage will be fed into the motor. To protect your motor, you should always band limit your signal (especially derivative feedback) to a value within 50 Hz.

■ **Caution:** Input ± 27 V, 4 A peak, 2.5 A continuous.

■ **Caution:** Exposed moving parts.

2.2.2 Road Simulation Brushed Servo Motor (Component # 17)

The Active Suspension incorporates a **Magmotor Brushed Servo Motor (S23-100 series)**, as shown in Figure 2.3b by ID #17. The motor has a power of 70 Watts. The motor is connected to the lead screw through the geared pinion and cable transmission. The lead screw translates the rotational motion of the motor to linear motion of the bottom plate. Some of the motor specifications are included in Figure 2.4. More detailed motor specifications are available in Reference [3].

FRAME SIZE	STACK LENGTH	PEAK STALL TORQUE (T_p) OZ-IN	CONT. STALL TORQUE (T_c) OZ-IN	ROTOR INERTIA (J_w) OZ-IN-SEC ²	FRICTION TORQUE (T_f) OZ-IN	THERMAL RESISTANCE (R_M) °C/WATT	MAX RECOMMEND SPEED RPM	MAX WINDING TEMP. °C	POWER RANGE W	WEIGHT LB
S23	-- 100	500	50	0.006	5	4.5	4000	155	70	2

Figure 2.4: Brushed Servo Motor Characteristics

■ **Caution:** *High-frequency signal applied to a motor will eventually damage the gearbox motor and the motor brushes.* The most likely source for high frequency noise is derivative feedback. If the derivative gain is set too high, a noisy voltage will be fed into the motor. To protect your motor, you should always band limit your signal (especially derivative feedback) to a value within 50 Hz.

■ **Caution:** Input ± 27 V, 24 A peak, 3 A continuous.

2.2.3 Lead Screw (Component # 18)

The lead screw, shown by ID #18 in Figure 2.1, circulates through a ring that is attached to the bottom of the bottom plate, component #5 in Figure 2.1. It is rotated by the motor. The lead screw has a pitch of 0.50 inches. Thus the road simulation plate moves 0.50 inches, or 1.24 cm, per single ball-screw revolution.

2.2.4 Top Plate Encoder (Component # 34)

Digital position measurement of the top plate vertical motion is obtained by using a high resolution quadrature optical encoder. The encoder used is a US Digital S1 single-ended optical shaft encoder that offers a high resolution of 4096 counts per revolution in quadrature mode (1024 lines per revolution). The complete specification sheet of the S1 optical shaft encoder is given in [1]. The effective resolution, i.e. minimum linear position that can be detected, of the stage displacement is $4.87 \mu\text{m}$.

Pushing the top plate upwards should result in positive change of encoder reading. Remark that incremental encoders measure the relative angle of the shaft.

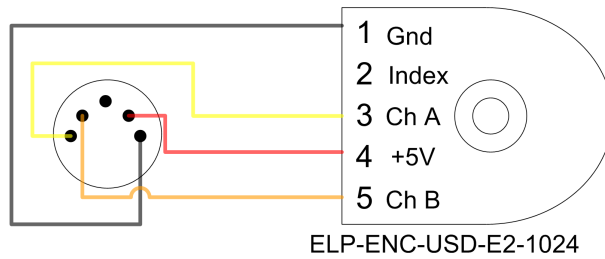


Figure 2.5: ASE top plate encoder wiring

The position signal generated by the encoder can be directly connected to the data-acquisition device using a standard 5-pin DIN cable. The internal wiring of the encoder and the 5-pin DIN connector on the ASE is illustrated in Figure 2.5.

■ **Caution:** Make sure you connect the encoder directly to your data-acquisition device and not to the power amplifier.

2.2.5 Suspension Encoder (Component #4)

Digital position measurement of suspension travel is obtained by using a high-resolution quadrature optical encoder. The optical encoder is directly mounted on the rear of the active suspension DC motor. The motor encoder has a resolution of 1000 lines per revolution. In quadrature mode this gives 4000 counts per full rotation of the encoder shaft. The effective resolution, i.e. minimum linear position that can be detected, of the stage displacement is $9.42 \mu\text{m}$. The encoder standard 5-pin DIN connector is pictured as component #32 in Figure 2.1. Pushing the middle plate and top plate towards each other should result in negative change of encoder reading.

2.2.6 Bottom Plate Encoder (Component #16)

Digital position measurement of bottom plate is obtained by using a high-resolution quadrature optical encoder. The optical encoder is directly mounted on the rear of the brushed servo motor (Component #17). The motor encoder has a resolution of 2048 lines per revolution. In quadrature mode this gives 4096 counts per full rotation of the encoder shaft. The effective resolution, i.e. minimum linear position that can be detected, of the stage displacement is $2.19 \mu\text{m}$. Refer to reference [2] for more details on this encoder. The encoder standard 5-pin DIN connector is pictured as component #11 in figure 2.1. Pushing the bottom plate upwards should result in a positive change of encoder reading.

2.2.7 Springs (Components #7 and #9)

The following sets of springs are supplied with the ASE as shown in Table 2.3.

Name	Application	Length	Quantity	Color	Stiffness per Spring
FUF-14-128-A	Suspension	0.128m	2	Black (installed upon delivery)	490N/m
FUF-14-128-B	Suspension	0.128m	2	Orange	1020N/m
FUF-14-115-A	Tire	0.115m	2	Black	570N/m
FUF-14-115-B	Tire	0.115m	2	Orange (installed upon delivery)	1250N/m

Table 2.2: Active Suspension Experiment Spring Nomenclature

Moreover, two springs are installed on the device upon delivery (Component #13). These springs are used to compensate for the weight of the three moving plates and reducing the pressure on the servo motor (Component #17).

2.2.8 Linear Bearing Block (Component # 10)

For smooth motion, the plates are fitted with low friction linear bearing blocks, shown by component #10 in Figure 2.1 , that glide on four stainless steel shafts, identified by component #20 in Figure 2.1.

2.2.9 Limit Switch (Components #24, 27, and 30)

The limit switch gets triggered when the bottom plate (Component #5) moves too close to its vertical range limits. In such a case this solenoid sensor directly deactivates the power of the servo motor (Component #17) and the red light will be on (see component #24 in Figure 2.1). In this situation the simulation should be stopped by the operator in the software and the bottom plate (Component #5) should be moved inside its working range. At this point the two red and green lights will be off, meaning that the plate is inside its working range but the motor is still not active. Then the limit switch push-key button (Component #27) should be pressed in order to activate the servo motor (Component #17) again. Table 2.3 describes different combinations of the limit switch lights.

Green Light	Red Light	Description
Off	Off	The bottom plate (Component #5) is inside its working range but the servo motor (Component #17) is not active. Push the limit switch button (Component #27) to activate the motor.
On	Off	The bottom plate (Component #5) is inside its working range and the servo motor (Component #17) is active.
Off	On	The bottom plate (Component #5) is outside its working range and the servo motor (Component #17) is not active. Follow these instructions: 1- Stop the simulation in the software. 2- Manually move the bottom white plate upwards with hand until the red LED on the bottom panel becomes lit. This is the upper position limit of the white plate. The motor power is cut off at this point (i.e. when the red LED is on.) 2- Manually move the bottom white plate downwards with hand until the red LED becomes lit. This is the lower position limit of the white plate. 3- Move the white plate to a location mid way from the two limit switches. 4- Press on the limit push button. The green LED should be lit. When lit, the power to the bottom motor is enabled.
On	On	The limit switch circuit is not working properly. Contact Quanser's technical support.

Table 2.3: ASE Limit Switch Lights Description.

2.2.10 Accelerometer (Component #21)

A dual-axis ADXL210E accelerometer is mounted underneath the top blue plate (Component #1) to measure the acceleration of the vehicle body along vertical direction. It is shown with ID #21 label in Figure 3. The accelerometer has the capability to measure both AC/dynamic accelerations (e.g. vibrations) or DC/static accelerations (e.g. gravity). The arrows represented on the accelerometer, and depicted in Figure 3, show the positive direction of the acceleration sensor measurement on its axes of sensitivity. To best measure the ASE top plate (vehicle body) vibration, the accelerometer is mounted such that its x-axis is aligned with vertical direction. Tapping the top plate (component #1) upwards should result in an initial negative measured acceleration voltage. In order for the accelerometer reading to be consistent with the encoder measurements its sign should be negated in the control software. The sensor has a range of ± 10 g and its noise, in the operating range of the top plate, is approximately ± 5.0 mV, i.e. ± 5.0 mg. This analog sensor is calibrated such that 1 Volt equals 1 g, or 9.81 m/s². Although the ASE accelerometer has already been calibrated at the factory, the signal conditioning circuit properties may vary depending on the external conditions (e.g. humidity, temperature). Therefore, you may want to adjust the Offset potentiometer (shown as component #2 in Figure 3) such that it reads approximately zero Volts with zero acceleration (i.e. sensor resting flat/horizontally). In a vertical configuration, the sensor should read -1 Volts.

■ **Caution:** The accelerometer readings can be misleading and lead to unexpected results. Please use caution when using them for control purposes as they are generally used as indicators.

3 ACTIVE SUSPENSION EXPERIMENT SPECIFICATIONS

Table 3.1 lists and characterizes the main parameters (e.g. mechanical and electrical specifications, conversion factors) associated with the Active Suspension Experiment. Some of these are used for the mathematical modeling of the plant as well as to obtain the system Equations Of Motion (EOM).

Symbol	Description	Value	Variation
	Structure Total Height	0.53 m	
M_s	Top Plate (Blue) with Attached Equipment Total Mass	2.45 kg	12%
M_{us}	Middle Plate (red) with Attached Equipment Total Mass	1 kg	12%
K_s	Suspension Passive Linear Stiffness Constant	Refer to Section 2.2.7	
K_{us}	Tire Linear Stiffness Constant	Refer to Section 2.2.7	
B_s	Inherent Suspension Damping Constant	$7.5 \text{ N-s}^2/\text{m}$	12%
B_{us}	Inherent Tire Damping Constant	$5 \text{ N-s}^2/\text{m}$	12%
	Suspension Motor Torque Constant	0.115 N-m/A	
	Suspension Motor Shaft Radius	0.006 m	
	Suspension Encoder Resolution	$942 \times 10^{-6} \text{ m/count}$	
	Bottom (White) Plate Encoder Resolution	$219 \times 10^{-6} \text{ m/count}$	
	Middle (Red) Plate Encoder Resolution	$487 \times 10^{-6} \text{ m/count}$	
	Suspension Travel Range	$3.80 \times 10^{-2} \text{ m}$	
	Middle (Red) Plate Travel Range	$3.50 \times 10^{-2} \text{ m}$	
	Bottom (White) Plate Travel Range	$3.60 \times 10^{-2} \text{ m}$	
	Accelerometer Sensitivity	$9.81 \text{ m/s}^2/\text{V}$	

Table 3.1: Main ASE Specifications

4 ENCODER AND FORCE GAIN CALCULATIONS

This section describes how the encoder count-to-position calibration gain (as shown in Table 4.1), as well as the force gain are calculated for the Active Suspension Experiment.

Encoder	Channel	Resolution (counts/rev)	Quadrature Mode Resolution (counts/rev)	Calibration Gain (m/count)
Top Plate Encoder (blue plate)	EI #2	1024	4096	4.8704×10^{-6}
Suspension Encoder	EI #0	1000	4000	9.425×10^{-6}
Bottom plate encoder (road)	EI #1	2048	8192	-2.197×10^{-6}

Table 4.1: Encoder gains

Top Plate

Encoder shaft radius = $0.25/2 = 0.125$ inch. Linear displacement found using the sector formula: $x = \theta \times r$, where $\theta = \text{angle from the encoder} = 2 \times \pi/4096$ and $r = 0.125 \times 25.4$ mm. Calibration gain (m/counts) is calculated as: $(2\pi/4096) \times (0.125 \times 25.4/1000) = 4.8704 \times 10^{-6}$ m/count.

Suspension

Encoder is mounted directly on active suspension motor. Motor shaft radius = 6 mm. Linear displacement found using the sector formula: $x = \theta \times r$, where $\theta = \text{angle from the encoder} = 2 \times \pi/4000$ and $r = 6$ mm. Calibration gain (m/counts) is calculated as: $(2\pi \times 6)/4000/1000 = 9.425 \times 10^{-6}$ m/count.

Bottom plate

The encoder is directly mounted on the rear of the brushed servo motor. The lead screw pitch is 0.5 inch/rev and coupled this with the gear ratio of the belt, every revolution of the encoder shaft means the bottom plate has moved 18 mm (or 0.018 m). Calibration gain (m/counts) is calculated as: $18/8192/1000 = -2.197 \times 10^{-6}$ m/count.

Force gain

Current-torque constant of active suspension motor is $k_t = 0.115$ N-m/A. Motor shaft radius = 6 mm. Force gain is calculated as: $0.115/0.006 = 19.2$ N/m.

Max force

Using a maximum current of 2A (approximately), the maximum force is calculated as: $19.2 \times 2 = 38.3$ N.

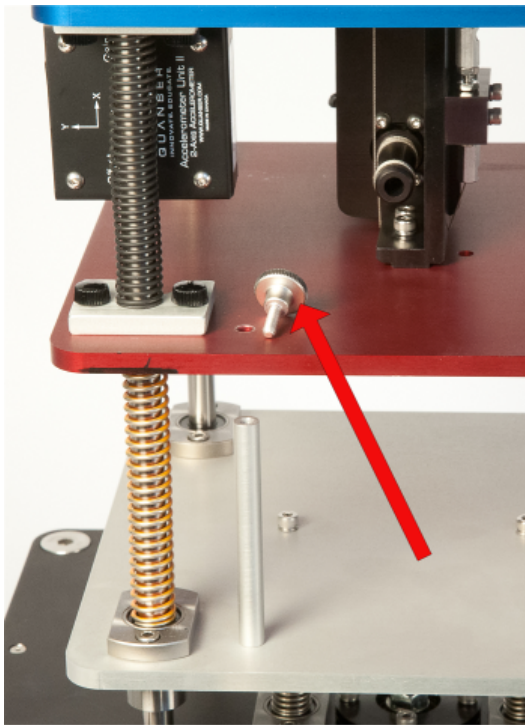
5 SETUP AND CONFIGURATION

This section describes the standard procedure to safely set up the mechanical components of the Active Suspension Experiment. Follow the steps described below to set up the Active Suspension Experiment mechanical assembly.

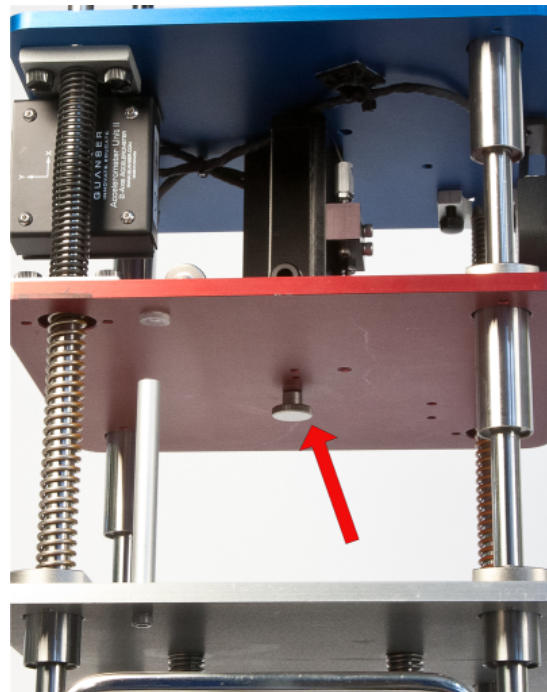
■ **Caution:** If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.

5.1 Unpacking and Routine Checkup

1. Cut the tie wraps that lock the bottom plate (component #5) to the plant handles (Component #25).
2. Unscrew and take off the two safety screws as depicted with red arrows in Figure 5.1. One screw is used to connect the rod (Component #28) to the red plate (Component #3) and thus keeps the white and red plates attached to each other. The other screw connects the red plate to the capstan holder so that the red and blue plates are attached to each other.
3. Ensure that the linear bearings (Component #10) are well lubricated. To test this, move the bottom plate (Component #5) with your hand. This should result in smooth vertical motion of the three plates (Component #1, #3, and #5) on the linear bearings. If not, a most probable cause of this issue is that the spring holders (Component #29) are in touch with the shafts. Refer to Section 4.2 to adjust the linear bearings and fix this problem.



(a) Safety screw for tightening the middle and bottom plate together before packing.



(b) Safety screw for tightening the middle and top plate together before packing.

Figure 5.1: Safety Screws.

4. Make sure that the top black cover plate (Component #23) is tightened with the four screws and the middle plate encoder cable (Component #19) is wrapped around the encoder shaft under enough tension. Refer to Section 4.3 for more details about encoder cable.

5. Before starting the controller, (assuming the system is wired properly), make sure the bottom plate (Component #5) is inside its working range by investigating the Component #30 in Figure 5 and the lights in Figure 2 (green should be On and red should be Off).
6. Clear the plates workspace from anything other than the intended components.

5.2 Changing the Springs

If the tire or suspension springs need to be changed then follow the steps below

1. Open the two screws on the top cover plate (component #23) as shown in Figure 5.2. The two holes in the top cover are designed for replacing the springs. Take out the rings from the two holes.

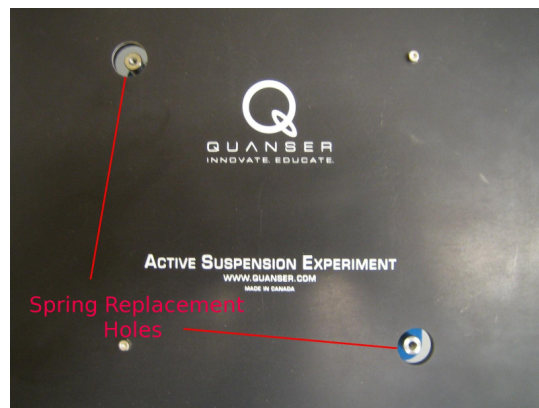
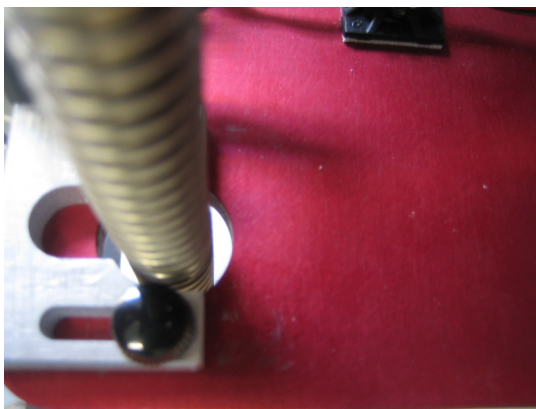
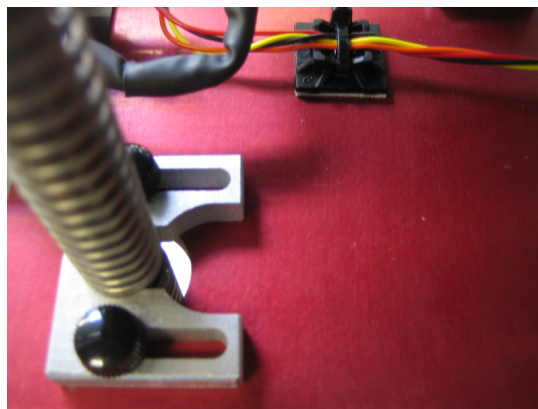


Figure 5.2: Active Suspension Experiment Top View

2. Loosen the screws on the four movable spring holders (component #29) to the degree that one can freely slide the holders on the plates. It should be mentioned that opening the screws for half a turn should be enough to loosen them and to disengage the spring holders. See Figure 5.3. **For each spring to be replaced there are two spring holders on the corresponding shaft that have to be moved.**



(a) Spring Holder in Open Mode.



(b) Spring Holder in Holding Mode.

Figure 5.3: Spring Holders.

3. Slide the spring holders (component #29) on the plates carefully until the spring pops out on the shaft. See Figure 5.3.

4. Replace the springs with the desired ones. Make sure that the springs representing the tire stiffness stand between the bottom plate (component #1) and the middle plate (component #3). Also, make sure that the springs representing the vehicle suspension stiffness stand between the middle plate (component #3) and the top plate (component #5).
5. Slide the four spring holders (component #29) until the springs are in contact with them and held well. Tighten the screws on the spring holders. **Make sure that all the four movable spring holders (component #29) are tightened and are not engaged with the shafts.**

5.3 Installing the Top Cover Plate and the Middle Plate Encoder

In case the middle plate encoder thread or the structure top cover plate come off, follow these steps to install them back.

1. There are four screw holes on the top cover plate (Component #23). Align each hole near the center of their corresponding shaft (Component #20). Use an Allen key and the four screws to tighten the screws and fasten the top cover to the plant structure.
2. Wrap the encoder thread (Component #19) around the top plate encoder shaft (Component #34). The rotation around the encoder shaft should be clock-wise when facing it. As a result, pushing the top plate (Component #3) upwards will result in a positive change in the encoder reading.
3. Lock the end of the thread in the anchor (Component #33). The number of turns on the encoder shaft should be such that the thread (Component #19) is neither loose nor highly tensioned.

6 WIRING PROCEDURE

The following is a listing of the hardware components used in this experiment:

1. **Power Amplifier:** Quanser AMPAQ, or equivalent.
2. **Data Acquisition Board:** Quanser Q1-cRIO, QPID/QPIDe, or equivalent.
3. **Active Suspension Experiment**

See the corresponding documentation for more information on these components. The cables supplied with the ASE are described in Section Section 6.1 and the procedure to connect the above components is given in Section 6.2.

6.1 Cable Nomenclature

The cables used to connect the Quanser Active Suspension Experiment with a power amplifier, analog sensors adapter, and data acquisition device are shown in Table 6.1.

Cable	Type	Description
 (a) RCA Cable	2xRCA to 2xRCA	This cable connects an analog output of the data acquisition terminal board to the power module for proper power amplification.
 (b) Motor Cable	5-pin-DIN to 4-Pronged male	This cable connects the output of the power module, after amplification, to the desired DC motor on the plant. It has a 4-pronged male connector at one end and a 5-pin-DIN connector at the other end. Each of these connectors are keyed so that they can be plugged in only one way. This cable has a rotating cover at its 4-pronged male connector that allows it to be screwed securely to the power module. (2 supplied)
 (c) Encoder Cable	5-pin-stereo-DIN to 5-pin-stereo-DIN	This cable carries the encoder signals between an encoder connector and the data acquisition board (to the encoder counter). Namely, these signals are: +5 VDC power supply, ground, channel A, and channel B

Table 6.1: Cables used to connect ASE to amplifier and data acquisition (DAQ) device

6.2 Typical Connections

This section describes the typical connections used to connect the ASE to a data-acquisition board and a power amplifier. The connections are summarized in Table 6.2, and pictured in Figure 6.1. Connection details are given below.

Cable	From	To	Signal
1	DAQ Analog Output #1	Amplifier <i>input #1</i> connector	Active Suspension DC Motor Amplifier Command.
2	DAQ Analog Output #0	Amplifier <i>input #0</i> connector	Base Motor Amplifier Command.
3	DAQ Analog input #0	ASE Accelerometer X connector	Carries the x-axis (vertical axis) accelerometer reading.
4	DAQ Encoder Input #2	ASE top plate <i>Encoder</i> connector	Top plate encoder angle measurement.
5	DAQ Encoder Input #1	ASE base <i>Encoder</i> connector	Base plate encoder angle measurement.
6	DAQ Encoder Input #0	ASE <i>Motor Enc</i> connector	Suspension angle measurement.
7	Amplifier: Output #0	ASE suspension <i>Motor</i> connector	Power leads to the active suspension dc motor.
8	Amplifier: Output #1	ASE base <i>Motor</i> connector	Power leads to the ASE base motor.

Table 6.2: Active Suspension Experiment Wiring Table.

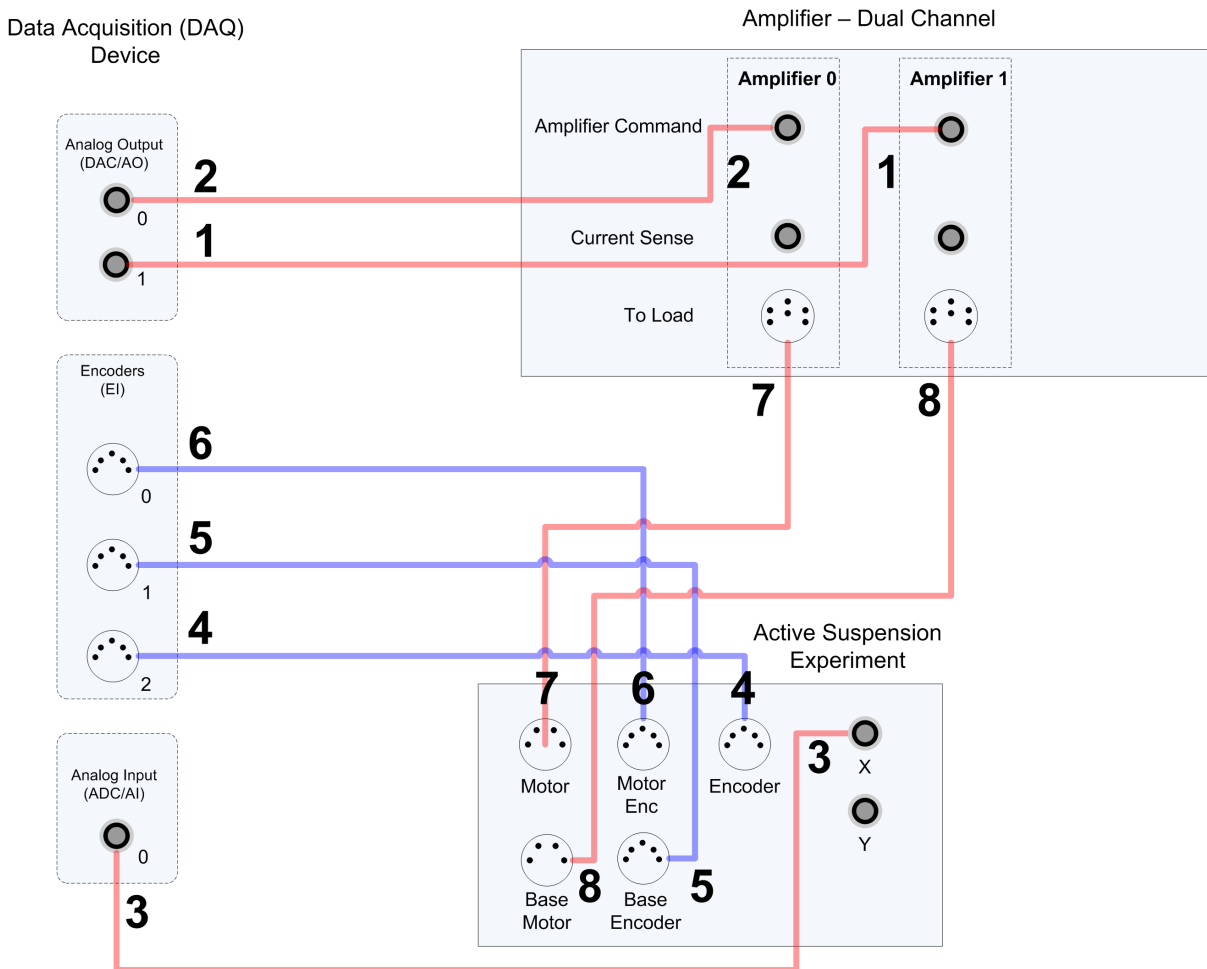


Figure 6.1: Connecting the Active Suspension Experiment to a Double-Channel Amplifier and a data acquisition (DAQ) device with 3 Encoder channels, 2 Analog Output channels, and 1 Analog Input channel.

Follow the steps below to wire the Active Suspension Experiment:

Note: The wiring diagram shown in Figure 6.1 is using a generic data acquisition device. The same connections can be applied for any data acquisition system that has at least 1x analog input channel, 2x analog output channels, and 3x encoder input channels.

1. Using the RCA to RCA cable, connect **Analog Output Channel #0** on the data-acquisition (DAQ) device to the **Input #0** connector on the amplifier. Also, connect **Analog Output Channel #1** on the data-acquisition (DAQ) device to the **Input #1** connector on the amplifier.
2. Using the RCA to RCA cable, connect **Analog Input Channel #0** on the data-acquisition (DAQ) device to the **X** accelerometer connector on the ASE.
3. Using the 5-pin-stereo-DIN to 5-pin-stereo-DIN encoder cable, connect the **Encoder #0** (ENC #0) on the data-acquisition (DAQ) device to the **ENCODER** connector on the top panel of ASE. Connect the **Encoder #1** (ENC #1) on the data-acquisition (DAQ) device to the **ENCODER** connector on the bottom panel of ASE. Connect the **Encoder #2** (ENC #2) on the data-acquisition (DAQ) device to the **MOTOR ENC** connector on the top panel of ASE.
4. Using the 3-pin-DIN to 4-pin-DIN motor cable, connect the **Output #0** on the amplifier to the **MOTOR** connector on the top panel of ASE. Also connect the **Output #1** on the amplifier to the bottom panel of **MOTOR** connector on the ASE. The 3-pin-DIN side of the motor cable should be connected to the amplifier side by first inserting the connector and then rotating the outer ring until it properly latches in place.

7 TESTING AND TROUBLESHOOTING

This section describes some functional tests to determine if your Active Suspension Experiment is operating normally. It is assumed that the ASE is connected as described in the Section 6, above. To carry out these tests, it is preferred that the user use a software such as **QUARC®** or **LabVIEW™** to read sensor measurements and feed voltages to the motor. Alternatively, these tests can be performed with a signal generator and an oscilloscope.

7.1 Active Suspension Motor

7.1.1 Testing

Ensure the active suspension motor is operating correctly by going through this procedure:

- Apply a voltage to analog output channel #0 of the data acquisition (DAQ) board.
- The two top plates should get farther apart from each other by applying the voltage.

7.1.2 Troubleshooting

If the motor is not responding to a voltage signal, go through these steps:

- Verify that the power amplifier is functional. For example when using the Quanser AMPAQ device, is the green LED lit?
- Check that the data-acquisition device is functional, e.g. ensure it is properly connected, that the fuse is not burnt.
- Make sure the voltage is actually reaching the motor terminals (use a voltmeter or oscilloscope).
- If the motor terminals are receiving the signal and the motor is still not turning, your motor might be damaged and will need to be repaired. Please see Section 8 for information on contacting Quanser for technical support.

7.2 Base Motor

7.2.1 Testing

Ensure the base motor is operating correctly by going through this procedure:

- Apply a voltage to analog output channel #1 of the terminal board using.
- The bottom plate should start moving upwards until it reaches the limit switch. See Section 2.2.9.

7.2.2 Troubleshooting

If the motor is not responding to a voltage signal, go through these steps:

- Verify that the power amplifier is functional. For example when using the Quanser AMPAQ device, is the green LED lit?

- Check that the data-acquisition board is functional, e.g. ensure it is properly connected, that the fuse is not burnt.
- Make sure the voltage is actually reaching the motor terminals (use a voltmeter or oscilloscope).
- If the motor terminals are receiving the signal and the motor is still not turning, your motor might be damaged and will need to be repaired. Please see Section 8 for information on contacting Quanser for technical support.

7.3 Encoders

7.3.1 Testing

Follow this procedure to test the ASE encoders:

1. Measure Encoder Input Channels #0, #1, and #2 on the data acquisition (DAQ) device in the software. These channels correspond to suspension encoder, base motor encoder, and top plate encoder, respectively.
2. Moving the blue and silver plates upwards should result in positive reading in top and bottom plate encoder values. In case of the suspension encoder, increasing the distance between the two top plates should result in positive reading in the corresponding encoder.

Note: Some data acquisition systems do not measure in quadrature mode and, in this case, one-quarter of the expected counts are received, i.e. 1024 counts in the ASE. In addition, some data acquisition systems measure in quadrature but increment the count by 0.25 (as opposed to having an integer number of counts). Make sure the details of the data-acquisition system being used is known. The counters on the Quanser DAQ boards measure in quadrature and therefore a total of four times the number of encoder lines per rotation, e.g. a 1024-line encoder results in 4096 integer counts for every full rotation.

7.3.2 Troubleshooting

If the encoder is not measuring properly, go through this procedure:

- Check that the data-acquisition (DAQ) device is functional, e.g. ensure it is properly connected.
- Check that both the A and B channels from the encoder are properly generated and fed to the data-acquisition device using an oscilloscope. There should be two square waves, signals A and B, with a phase shift of 90 degrees. If this is not observed then the encoder may be damaged and need to be replaced. Please see Section 8 for information on contacting Quanser for technical support.

7.4 Accelerometer

7.4.1 Testing

Follow this procedure to test the ASE accelerometer:

- Measure Analog Input Channel #0 on data-acquisition (DAQ) device in software. This channel corresponds to ASE Accelerometer X connector.
- Tapping the top plate upwards should result in positive spike signal measured by the accelerometer.

7.4.2 Troubleshooting

Follow the steps below if the accelerometer is not measuring correctly::

- Check that the data-acquisition (DAQ) device is functional, e.g. ensure it is properly connected.
- Measure the voltage across the accelerometer output. If the voltage from the accelerometer does not change when you move the top plate, your accelerometer needs to be replaced. Please see Section 8 for information on contacting Quanser for technical support.

8 TECHNICAL SUPPORT

To obtain support from Quanser, go to <http://www.quanser.com/> and click on the Tech Support link. Fill in the form with all the requested software and hardware information as well as a description of the problem encountered. Also, make sure your e-mail address and telephone number are included. Submit the form and a technical support person will contact you.

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- [1] US Digital. *E2 Optical Kit Encoder*, 2007.
- [2] US Digital. *USDigital-E3-Optical Kit Encoder*, 2007.
- [3] Magmotor. *Magmotor-S23-Features*, 2007.

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